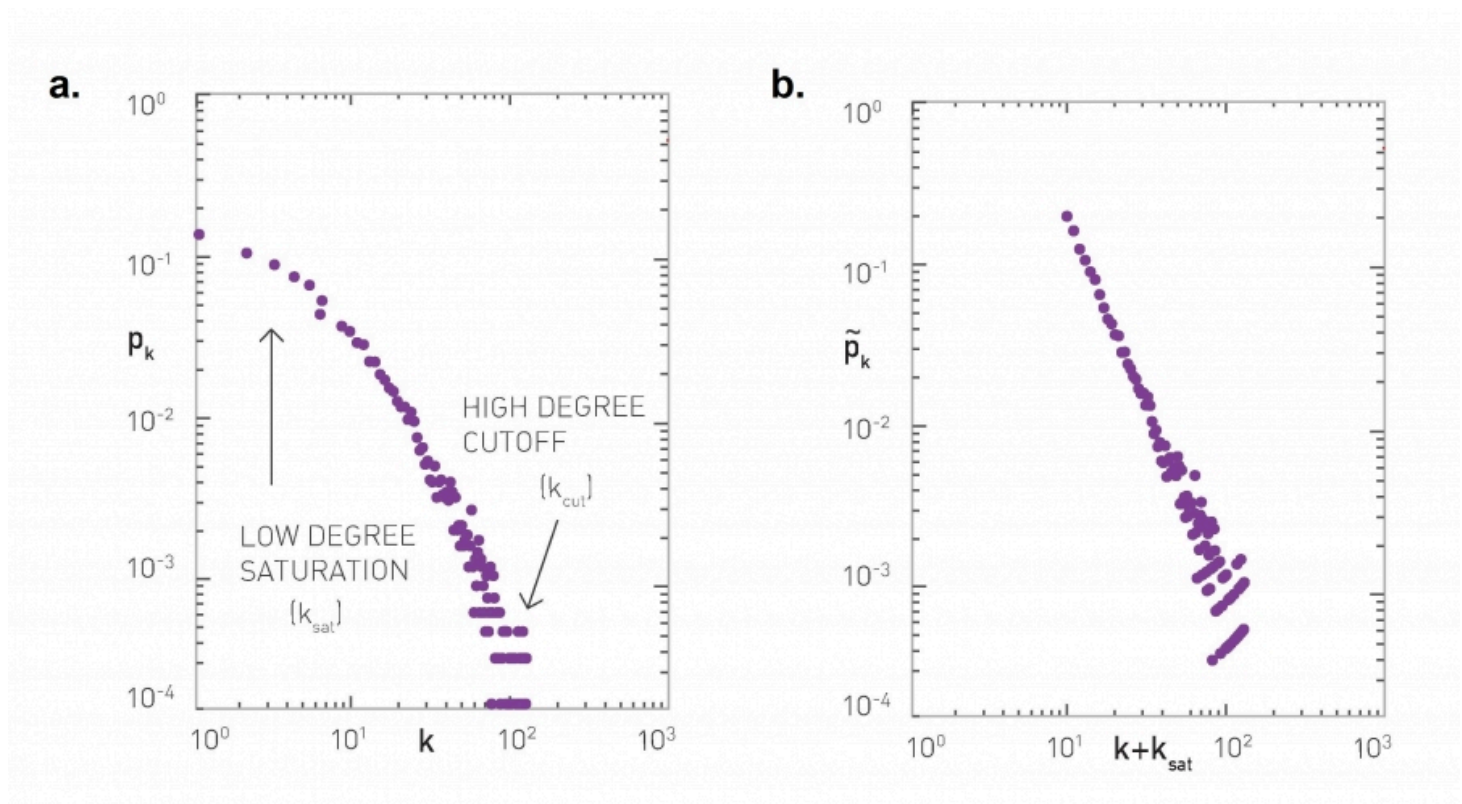


L4: How to Check if a Network Has Power-law Degree Distribution



Rescaling the Degree Distribution, Figure 4.23 from networksciencebook.com by Albert-László Barabási

In practice, the degree distribution may not follow an ideal power-law distribution throughout the entire range of degrees. In other words, the empirical degree distribution may not be a perfect straight-line when plotted in log-log scale. Instead, it may look similar to the left plot at this page. There are two important points about this distribution:

a) For lower values of k , we observe a “low-degree saturation” that decreases the probability of seeing low-degree nodes compared to an ideal power-law. If this saturation effect takes place mostly for nodes with degree less than k_{sat} , we can capture this effect by considering a modified power-law expression: $(k + k_{sat})^{-\alpha}$

Do you see why adding the term k_{sat} causes a decrease in the probability of low-degree nodes (but it has a minor effect on high-degree nodes)?

b) For very high values of k there is again a deviation from the ideal power-law form. The reason is that in practice there are always some “structural” effects that limit the maximum degree that a node can have. For example, in a router-level computer network, the maximum degree is limited by the maximum number of interfaces that a router can have due to hardware or cost constraints. To capture such a high-degree cutoff point, we often need to constrain the upper-tail of the degree distribution up to a value k_{cut} . This value is determined by practical connectivity constraints for the network we analyze.

In summary, when we want to check if a network follows a power-law degree distribution, we need to consider whether that distribution drops almost linearly with k in a range between k_{sat} and k_{cut} .

Another approach, that addresses the low-degree saturation effect, is to “rescale” the distribution so that we examine the behavior of the degree probability with $(k + k_{sat})$ (instead of k) – still considering the range up to k_{cut} . This rescaling approach is shown at the plot at the right.

Food for Thought

A more rigorous statistical approach to examine if a network has a power-law degree distribution is described at the following site: [Power-law Distributions in Empirical Data](https://aaronclauset.github.io/powerlaws).  <https://aaronclauset.github.io/powerlaws>. You will experiment with this (or similar) method in an assignment. For now, you may just want to read the description of that method from the given URL.